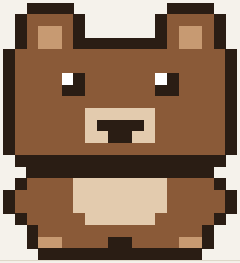
ROMA TRE · A.Y. 2026/27

# Policy Evaluation Lab

*Lecture 5: Randomised experiments (II)*



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# Where we are

- Lecture 4: a lottery balances observed **and** unobserved characteristics, so the difference in means is the causal effect
- Real experiments are messier than a coin toss

Today:

1. what can go wrong: firms that do not take up the treatment, firms that drop out, spillovers, samples that are too small
2. a real experiment on firms: does **exporting** make firms better? (Atkin, Khandelwal and Osman, 2017)
3. experiments inside companies: **A/B tests**

PART 1

# Non-compliance



# Offering is not treating

- In most experiments with firms we can randomise the **offer**, not the treatment itself
  - a grant: some winners never invest (no co-financing, the project changes)
  - consulting: some firms do not implement the advice
  - an export order: some firms do not accept it
- Some control firms may obtain the treatment anyway, from other sources

Two variables, not one:

- $Z_i$: **assignment** (random): 1 if the firm won the lottery
- $D_i$: **take-up** (chosen by the firm): 1 if it actually received the treatment

# Do not compare takers with non-takers

- Tempting: compare winners who took up the treatment with everybody else
- But take-up is a **choice**: the winners who invest are the ones with a good project, a bank willing to lend, a motivated owner…
- We would be back to comparing Alfa with Beta: **selection bias**

> Always compare groups as they were **randomised**: all winners vs all losers, whatever they did afterwards

# Intention-to-treat (ITT)

$$ITT = Avg[Y_i \mid Z_i = 1] - Avg[Y_i \mid Z_i = 0]$$

- The effect of **being offered** the treatment
- Unbiased, because $Z_i$ is random
- Often the policy-relevant number: an agency can offer grants, it cannot force firms to invest
- But it **dilutes** the effect: winners who did nothing are averaged in

# Treatment on the treated (TOT)

If losers cannot get the treatment, divide the ITT by the share of winners who took it up:

$$TOT = \frac{ITT}{Avg[D_i \mid Z_i = 1] - Avg[D_i \mid Z_i = 0]} = \frac{\text{effect of the offer}}{\text{effect of the offer on take-up}}$$

**EXAMPLE: OUR GRANT LOTTERY**

419 lottery winners, but only 60% invest; the true effect on investors is 3 points:

- ITT $= 0.60 \times 3 = 1.8$ points (effect of the offer); TOT $= 1.8/0.60 = 3$ points (effect on investors)

The TOT is an instrumental-variables estimate, with the lottery as the instrument: more in the IV weeks.

PART 2

# Other threats



# Attrition

- Firms **drop out** of the study: they close, move, stop answering the survey
- Random attrition only reduces the sample
- **Differential** attrition creates bias: e.g. the weakest control firms close and disappear from the data, while the grant keeps weak treated firms alive $\Rightarrow$ the surviving controls look artificially good

What to check:

- attrition rates in treatment and control: are they similar?
- characteristics of those who leave: are they similar in the two groups?

# Spillovers

- The comparison works only if the treatment of one firm does not affect the control firms
- Firms interact: they compete for the same customers, copy each other, share suppliers and workers
  - **positive** spillovers: control firms learn from treated neighbours $\Rightarrow$ the effect is underestimated
  - **negative** spillovers: subsidised firms steal customers from control firms $\Rightarrow$ the effect is overestimated
- Bloom et al.: owners copied the practices to their other plants, so treatment and control plants were never in the same firm

> Solution: randomise at a higher level (firm, town, market)

# Power: two ways to be wrong

Before running an experiment, ask: **if the programme works, will we see it?**

| | **No true effect** | **True effect** |
|---|---|---|
| We find a significant effect | false positive (prob. $\alpha$) | correct |
| We find no significant effect | correct | false negative (prob. $\beta$) |

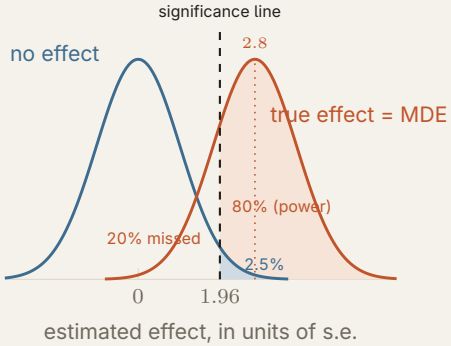
- $\alpha$ = significance level, chosen by us: usually **5%**
- **Power** $= 1 - \beta$: the probability of detecting an effect that is really there. Convention: **80%**

> **Minimum detectable effect (MDE)**: the smallest true effect that we would find significant at 5% in 80% of experiments like ours

# Where the MDE comes from

- Repeat the experiment many times: the estimates form a bell curve, with spread = **s.e.**
- **No effect**: centred on 0. Significant only above $1.96 \times s.e.$ (5%, two-sided)
- **True effect = MDE**: for 80% of the curve to clear the line, its centre must sit $0.84 \times s.e.$ **beyond** it

$$MDE = (1.96 + 0.84)\, s.e. \approx \mathbf{2.8}\ \mathbf{s.e.}$$



# The MDE formula

With $N$ units in total, a share $P$ of them treated, and outcome standard deviation $\sigma$:

$$MDE = \underbrace{(z_{1-\alpha/2} + z_{1-\beta})}_{1.96+0.84=2.8} \times \underbrace{\sigma\sqrt{\frac{1}{P(1-P)\,N}}}_{s.e.} \xrightarrow{P=1/2} MDE \approx \frac{5.6\,\sigma}{\sqrt{N}}$$

- **More units**: the MDE falls with $\sqrt{N}$: to halve it, we need **four times** the sample
- **Less noise**: a smaller $\sigma$ (similar firms, good measurement, controlling for the pre-treatment outcome) shrinks the MDE
- **Balanced groups**: $P = 1/2$ is best; treating only 20% of the firms raises the MDE by 25%

Stricter standards cost sample: with 90% power the multiplier is $1.96 + 1.28 = 3.24$ instead of 2.8.

# Power: how many firms do we need?

Turn the formula around to get the sample size:

$$N = \frac{2.8^2\,\sigma^2}{P(1-P)\,MDE^2} \qquad \text{3 points, } \sigma = 9.1: \quad N = \frac{7.84 \times 82.8}{0.25 \times 9} \approx 290 \text{ firms}$$

| **Firms in the lottery** (half winners) | **s.e.** | **MDE (points)** |
|---:|---:|---:|
| 100 | 1.82 | 5.1 |
| 200 | 1.29 | 3.6 |
| 400 | 0.91 | 2.6 |
| 1,500 | 0.47 | 1.3 |

Sales growth in our simulated firms: $\sigma = 9.1$ points.

A lottery among 100 firms would most likely **miss** our 3-point effect: about 300 firms are needed. “Not significant” in a small experiment does not mean “no effect”

# Power and non-compliance

- What we estimate directly is the **ITT**, so the ITT is what must exceed the MDE
- With take-up $c$, the ITT is only $c\times$ the effect on those treated: the sample needed grows with $1/c^2$

**OUR GRANT LOTTERY: 60% OF WINNERS INVEST, TRUE EFFECT ON INVESTORS 3 POINTS**

- ITT $= 0.60 \times 3 = 1.8$ points
- $N = \dfrac{7.84 \times 82.8}{0.25 \times 1.8^2} \approx 800$ firms, not 290: **almost three times as many**

- Randomising whole towns or stores (against spillovers) also costs power: what counts is the number of **groups**, not of firms

# External validity

- **Internal validity**: is the estimate a causal effect for the firms in the experiment? Randomisation is the best guarantee
- **External validity**: does it hold for other firms, places, times, scales?
  - firms that agree to an experiment are not a random sample of all firms (Bloom et al.: 17 of 66)
  - an effect for small Egyptian weavers may not hold for Italian manufacturers
  - a pilot on 50 firms may not scale: at national level, prices, competition and the agency's capacity change

PART 3

# Learning by exporting



# The question

- Exporters are larger, more productive and pay higher wages than non-exporters: one of the most robust facts in international economics
- Two explanations:
  - **selection**: only the best firms can pay the costs of entering foreign markets (Alfa exports, Beta does not)
  - **learning by exporting**: selling to demanding foreign buyers makes firms better
- The answer matters for policy: export promotion agencies (in Italy, ICE–Agenzia), trade fairs, export credit (SACE, SIMEST)

> With observational data we cannot separate the two: we need firms that export **by chance**

# The experiment

Atkin, Khandelwal and Osman (QJE, 2017): small **rug producers** in Fowa, Egypt

- Typical firm: one owner, about one worker, a hand loom; only about 12% had ever knowingly produced for export
- A US NGO (Aid to Artisans) found buyers in high-income countries (e.g. a German importer); a local intermediary handled the orders
- **Treatment**: the intermediary offered an export order of 110 $m^2$ of rugs, about **11 weeks of work**; further orders followed if the buyer was satisfied
- **Control**: not contacted
- Randomisation within **strata** (rug type and loom size)

# Take-up: the offer matters

| | Firms | Offered | Took up |
|---|---|---|---|
| Sample 1 (2011): small orders, 10–20 $m^2$ at a time | 79 | 39 | 14 |
| Sample 2 (2012): full 110 $m^2$ order at once | 140 | 35 | 32 |
| **Joint sample** | **219** | **74** | **46** |

Producers of the main rug type (duble). Source: Atkin, Khandelwal and Osman, NBER WP 20690, Table 1.

- When orders were small and uncertain, few firms accepted (36%); with a real order, almost all did (91%)
- Overall take-up: $46/74 \approx 62\%$. Non-compliance is part of the design, so they report both ITT and TOT

# Balance and attrition

- **Balance**: no significant differences between treatment and control firms at baseline in owner age, experience, literacy, household income, profits, hours, output
- One exception: treatment firms had **lower** rug quality at baseline. Chance happens (Lecture 4), and it works against finding a quality improvement
- **Attrition**: about 11% of firms missing per survey round, the same in treatment and control
- **Spillovers**: no evidence that control firms close to treated firms were affected

> Every threat from the first part of the lecture is checked, and reported

# Results: exporting and profits

| | **ITT** (effect of the offer) | **TOT** (effect of exporting) |
|---|---|---|
| Ever exported (control mean 0.20) | $+0.55^{***}$ | $+0.76^{***}$ |
| | (0.06) | (0.07) |
| Monthly profits (log) | $+0.26^{***}$ | $+0.42^{***}$ |
| | (0.05) | (0.08) |

Source: Atkin, Khandelwal and Osman, NBER WP 20690, Tables 4 and 5 (direct profit measure). The published version reports profit gains of 16–26% across measures.

- Being offered an order raised profits by about **26%**; for the firms that exported, by about **42%**
- $0.42 / 0.26 \approx 1.6 \approx 1 / 0.62$: the TOT scales up the ITT by the take-up rate

# Results: learning by exporting

- Treatment firms produced **fewer** square metres per hour (24–28% fewer)
- But their rugs were of **higher quality** on almost all of 11 dimensions rated by a master artisan (corners, flatness, design accuracy…)
- A different product? At the end, all firms wove an **identical** domestic rug with the same inputs and loom: treatment firms made better rugs in the same time
- Quality rose most along the dimensions discussed with buyers and the intermediary

> Exporting made these firms **better**, not only richer: evidence of learning by exporting

PART 4

# Experiments inside the firm



# A/B tests

- Firms run randomised experiments every day: **A/B tests**
  - customers (or stores, or sales agents) are randomly shown version A or version B: a price, a web page, a promotion, an e-mail
  - the outcome is compared: purchases, clicks, revenue
- Large online firms run thousands of experiments a year
- Example: at Microsoft's Bing, a small change in how ad headlines were displayed, tested on randomly chosen users, raised revenue by 12% (Kohavi and Thomke, *Harvard Business Review*, 2017)

> An A/B test is an RCT: same logic, same potential outcomes, same difference in means

# The same threats, in a company

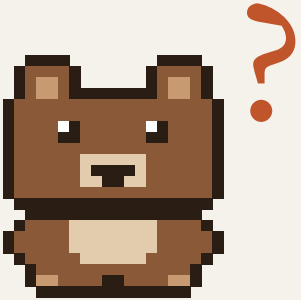
- **Non-compliance**: customers who receive a discount code but never use it ⇒ compare everyone who received it (ITT)
- **Spillovers**: a promotion in some stores draws customers from nearby control stores ⇒ randomise by city or region
- **Power**: small effects need huge samples: easy for a website, hard for 40 stores
- **External validity**: a pilot in the best-run stores, chosen by managers, is not an experiment: **randomise the pilot stores**

> **PILOTS AND ROLLOUTS**
>
> Roll out in random order: stores that start later are the controls for those that start earlier

# Lessons

- Randomise the offer, and analyse the groups **as randomised**: ITT, then TOT
- Check balance, attrition and spillovers, and report them
- Do the power calculation **before** the experiment: a small experiment can easily miss a real effect
- Experiments are the gold standard, but often impossible: governments rarely assign grants by lottery

**Next**: when we cannot randomise, can we find firms that are **comparable on observables**? Matching and the propensity score



# Questions?

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